1. You trained a k-means clustering model and it is highly sensitive to outliers in your data. Why?

Because the mean as a statistic is sensitive to outliers and k-means' centroid selection is also sensitive to outliers. ✓

2. Which best defines a high-quality dataset?

A dataset in which instances are complete, accurate, reliable, and relevant ✓

3. When using neural networks for multi-class classification, why will a single output node with `sigmoid` activation not work?

Because `sigmoid` on a single output node will give a single value between 0 and 1 and it is insufficient for multiple classes ✓

4. While exploring a dataset, you draw a scatterplot matrix of your observations to see how they distribute. There is a positive linear relationship in some of the plots, so you think that those features are somewhat correlated. What is the best way to confirm your assumptions?

Calculate the correlation coefficient between the presumed features. ✓

5. What is the primary assumption for using linear regression?

The relationship between dependent and independent variable is linear. ✓

6. What is the meaning of ensemble learning methods?

Using multiple learning methods to obtain better results ✓

7. What can you do about missing or incomplete data in a column that is relevant to train the model?

Use synthetic data to fill or replace values in the variable of concern. ✓

8. What does regularization mean?

Shrinking coefficients estimates toward zero and penalizing model complexity, to reduce the risk of overfitting ✓

9. You use a model in production to classify articles by topic, but it exhibits a drop in its prediction quality. You bring the model back to retrain it. After retraining the model with fresher data, what would you do next to describe the performance of the classification model?

Compute a confusion matrix. ✓

10. When the value of feature `x4` is more than 1,000, your model predictions are above the acceptance threshold; when its value less than 1,000, the model predictions are poor. What could be a reasonable way to fix this?

Aggregate the feature `x4` values that are less than 1,000 and refit the model based on the transformed data. ✓

11. What is a good method for identifying extreme values?

Through visualizations such as boxplots ✓

12. You are training a machine learning model over a dataset. While validating the model results, the model exhibits high error in both training and testing splits. What can you conclude from this?

The model is underfit. ✓

13. You are evaluating a neural network. The error diminishes through the epochs, but after a few epochs the error starts to increase. You want to apply early stopping on the epoch that shows a steady increase in error. Why is "early stopping" necessary for this model?

The learner is about to reach the overfit point and is no longer able to be generalized to new data. ✓

14. Which type of dependent variables do logistic regression models predict?

Categorical variables ✓

15. You have a recommender model for an online retailer that produces a list of items for the user as they browse the website. How can you ensure that this model continues to stay relevant with its recommendations?

By continuously tracking model performance on live data based on mean average precision, personalization, and coverage scores ✓

16. You want to tag news headlines with the categories they fall under. How will you select multiple categories using a single binary classification model?

Set a threshold on the output classes and select all categories over that threshold, in a one-vs-rest approach. ✓

17. You perform clustering analysis on a dataset of users based on their past behavior. How can you use the past data to validate the clusters?

Check the compactness of known reference items inside a cluster and their separation to other reference items from other clusters. ✓

18. What action is most likely to improve an underfitting model without changing the number of observations?

Adding additional features to the model ✓

1. While predicting the weight of a person in kilograms, a root-mean-square deviation (RMSE) of 50 is bad, but the same value while predicting the population of a city is excellent. Why is that?

RMSE values are directly dependent on the scale of the data. ✓

2. You are evaluating two logistic regression models: Model 1 and Model 2. You notice that Model 1 outputs a p-value of 0.76, whereas Model 2 outputs a p-value of 0.026 Which model will be more likely to perform better and why?

Model 2 because all features used for training display p-values that are lower than 0.05 ✓

3. In bagging, what method derives the final output from the outputs of N classifiers?

Average of all ✓

4. What kind of a clustering algorithm is density-based spatial clustering of applications with noise (DBSCAN)?

Density-based ✓

5. When should you use a 50/50 break to split testing and training sets?

- When the size of your dataset is large enough to accommodate different distributions in both the test and training data subsets
- When the distribution has a mean of 0 and a standard deviation of 1
- When the size of your dataset is large enough to accommodate similar distributions in both the test and training data subsets ✓

6. Reviewing the model error in comparison to the model complexity, you notice that the higher the complexity of the model, the higher its prediction error becomes. What action would you take next to improve the model?

Apply a logistic regression model.

Apply covariance analysis to identify highly correlated variables.

Apply regularization to the coefficients of the model. ✓

7. You are given a car's data where Y is miles per gallon (MPG) and X is horsepower (HP). If car A has HP = 15 and car B has HP = 20, what is the expected target value of A in comparison to B for the following regression equation?MPG = 30 – 0.7 HP

Car A is expected to have 3.5 miles per gallon more than Car B. ✓

8. How can you tell when a variable is contributing to the overall model performance in a linear model?

The coefficient for the variable has a p-value less than 0.05. ✓

9. You are building a machine learning model using a data generator tool such as Weka over a dataset. At first you train your model using all available features, but this overfits the model. How would you improve the model?

Perform feature selection in the data generator tool by choosing an attribute evaluator and a search method before training the model. ✓

10. What is an indicator that your model was trained on non-representative data?

The predicted values always exceed the mean of the data. ✓

11. When would you use a random forest to build a classifier for your data instead of a support vector machine (SVM)?

If the problem is a multiclass problem, the dataset doesn't fit in memory, or if the encoding of categorical values cannot be performed ✓

12. You are working on a machine learning model that a local government uses to track the price evolution of items in a basket of goods. The usual process consists of physically surveying citizens every quarter by asking for these item prices. What would you do to significantly improve the data collection process?

Scrape item prices from the producer/retail websites whenever possible. ✓

13. How can you ensure that given a fixed set of hyperparameters, k-means produces the same clusters on every run with the same dataset?

By setting the random seed value to a fixed value for all the iterations ✓

14. What is the primary assumption of using the extreme value method to identify outliers?

The data assumes a given distribution, usually Gaussian ✓

15. In a multi-graphics processing unit (GPU) architecture, how can you achieve data parallelism in a neural network?

Use the same weights but split the mini-batches to each GPU.

16. During principal component analysis (PCA)/factor analysis, the first factor has an eigenvalue of 3.87. The total of all the eigenvalues is 9. What proportion of variance is accounted for by this first factor?

43.00%

17. What is the purpose of bagging?

To improve the accuracy of the combined model while lowering its variance ✓

18. A poorly performing model with high bias and low variance is considered to be what?

Underfit ✓